
Iowa Liquor Database: SWOT Analysis SQL Questions

SQL

/* Name: [Your Name]

Category/Vendor of Choice: [e.g., CANADIAN WHISKIES / Jim Beam Brands]

Instructions: Copy and paste the below questions into your script.

Below each question, write your response, including relevant SQL queries.

Make sure the questions and plain-language answers are commented out!

*/

-- SELECT, Filtering & Sorting

-- 1. Create a list of all transactions for your chosen [Category/Vendor] that took place in
-- the last quarter of 2014, sorted by the total sale amount from highest to lowest.
-- (Strength: Identifying high-volume peak periods).

-- 2. Which transactions for your [Category/Vendor] had a bottle quantity greater than 12?
-- Display the date, store number, item description, and total amount.
-- (Strength: Identifying bulk buyers or wholesale-style transactions).

-- 3. Find all products in the products_table whose item_description contains a specific
-- keyword (e.g., 'Limited', 'Spiced'). What categories do they belong to?
-- (Opportunity: Identifying niche product variants).

-- Aggregation

-- 4. What is the total sales revenue and the average bottle price (btl_price) for
-- your chosen [Category/Vendor]?
-- (Strength/Baseline: Establishing the financial footprint).

-- 5. How many transactions were recorded for each specific item description within your
-- chosen [Category]? Which specific product is the most frequently purchased?
-- (Strength: Identifying your "hero" product).

-- 6. Which store generated the highest total revenue for your [Category/Vendor]?
-- Which generated the lowest (but still greater than zero)?
-- (Strength vs. Weakness: Identifying your best and worst retail partners).

-- 7. What is the total revenue for every vendor within your chosen [Category],

- sorted from highest to lowest?
 - (Threat: Identifying your top competitors in that space).
-
- 8. Which stores had total sales revenue for your [Category/Vendor] exceeding \$2,000?
 - (Hint: Use HAVING to filter aggregated store totals).
 - (Strength: Pinpointing high-performing retail locations).

-- Joins

- 9. Find all sales records where the category_name is either your chosen category or a closely related competitor category (e.g., 'VODKA 80 PROOF' vs 'IMPORTED VODKA').
 - (Threat: Comparing performance against direct substitutes).
-
- 10. List all transactions where the state bottle cost was between \$10 and \$20 for your [Category/Vendor].
 - (Opportunity: Analyzing performance in the "mid-tier" price bracket).
-
- 11. Write a query that displays each store's total sales for your [Category/Vendor] along with the store's name and address from the stores_table.
 - (Strength: Mapping your physical sales footprint).
-
- 12. For each sale in your [Category], display the transaction date, total amount, and the population of the county where the sale occurred by joining with counties_table.
 - (Opportunity: Correlating sales volume with population density).
-
- 13. Write a query that shows total sales for your [Category/Vendor] by county.
 - Which county generates the most revenue for you?
 - (Strength: Identifying your geographic stronghold).
-
- 14. Join the sales_table and products_table to show total revenue for your [Vendor] alongside the proof and pack size of the items.
 - (Strength: Determining if higher proof or larger packs drive more revenue).
-
- 15. Are there any counties in the counties_table that have a population over 10,000 but zero sales for your [Category/Vendor]?
 - (Opportunity: Identifying untapped markets).
-
- 16. Display total revenue for your [Category/Vendor] grouped by the store_status (from stores_table). Are active stores ('A') performing significantly better than others?
 - (Threat: Assessing the risk of sales tied to inactive or closed locations).

-- Subqueries

- 17. Using a subquery, find all transactions for your [Category/Vendor] from stores

- located in a specific high-growth city (e.g., 'Des Moines') found in the stores_table.
- (Opportunity: Drilling down into urban market performance).

- 18. Which stores had total sales for your [Category/Vendor] that were above the
- average store revenue for that same group? (Hint: Use a subquery for the average).
- (Strength: Identifying stores that are over-performing).

- 19. Find the top 5 highest-grossing items for your [Vendor], then use a subquery
- to look up their full details (like case_cost and profit) from the products_table.
- (Strength: Deep-dive into the specs of your most profitable inventory).

- 20. Write a query using a subquery to find all sales records for your [Category]
- from stores that have an 'A' (Active) status in the stores_table.
- (Strength: Filtering for reliable, ongoing revenue streams).